

Chapter 2 - Functions

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#Review

Ordered Pair	An ordered pair is a pair of objects whose components occur in a special (fixed) order. Two ordered pairs are equal only when their corresponding elements are equal. i.e., $(a, b) = (c, d) \Leftrightarrow (a = c) \wedge (b = d)$
Cartesian Product	A cartesian product is the set of all possible ordered pairs of two non-empty sets such that the first component and the second component are an element of the first set and second set respectively. Mathematically, $A \times B = \{(x, y) : x \in A, y \in B, A \neq \phi, B \neq \phi\}$
Relation	A relation is any subset of the cartesian product of two non-empty sets. Mathematically, $R: A \rightarrow B = \{(x, y) : x \in A, y \in B, A \neq \phi, B \neq \phi\} \subseteq A \times B$
Domain and Range of a Relation	Domain: Domain of a relation is the set of all 1st components of the ordered pairs of a relation. Mathematically, $Dom(R) = \{x : (x, y) \in R\}$ Range: Range of a relation is the set of all 2nd components of the ordered pairs of a relation. Mathematically, $Ran(R) = \{y : (x, y) \in R\}$

Inverse Relation	Inverse relation is the new relation obtained by interchanging all components of the ordered pairs of a relation. Mathematically, $R^{-1}: A \rightarrow B = R: B \rightarrow A = \{(y, x) : y \in B, x \in A, B \neq \phi, A \neq \phi\} \subseteq B \times A$
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#Function

Function is a relation from a set of inputs to a set of possible outputs where each input is related to exactly one output. In a function, the same input cannot give more than one output and the vertical line should not intersect the locus of a function at more than 1 point at any position.

A function from A to B is denoted by

$$f: A \rightarrow B \text{ or } A \xrightarrow{f} B$$

Terminology

Domain	The set of inputs that gives real output is called the domain of a function.	
Co-domain	The set of outputs is called the co-domain of a function.	
Range	The set of possible outputs is called the range of a function. Note: $Range \subseteq Co-domain$	
Pre-image	The input value is called pre-image.	If $f(a) = b$ then we can say: 1. a is the pre-image of b 2. b is the image of a
Image	The output value is called image.	

Equality of Two Functions

Two functions are equal if the domain and range of 1st and 2nd function are equal.

$$\text{Mathematically, } Dom(f) = Dom(g) \wedge Ran(f) = Ran(g) \Rightarrow f(x) = g(x)$$

Special Types of Function

1. Injection (Injective Function)

A function such that $f(x_1) = f(x_2) \Leftrightarrow x_1 = x_2$ or every distinct input has a distinct output is called a one to one or injective function.

2. Surjection (Surjective Function)

A function in which $Range = Co-domain$ or every image has at least one pre-image is called an onto or surjective function.

3. Bijection (Bijective Function)

A function which is both injective (one to one) and surjective (onto) is called bijective function.

#Inverse Function

Inverse function is the new function obtained by interchanging the elements of the domain and range of a bijective function. If a function maps set A to set B then its inverse function maps set B to set A .

Mathematically, $f: A \rightarrow B \Rightarrow f^{-1}: B \rightarrow A$

Property

The relation between a function and its inverse function is

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x$$

#Composition of Functions and Composite Function

Composition of functions means combining of functions or the process of applying one function after another function while composite function is the final function obtained after composition of functions.

Independently, Composite function is the new function that directly connects the first set to the last set for 2 or more than 2 non-empty sets.

Or, Composite function is the function that is formed when the output of one function is used as the input for another function.

If a function f maps set A to set B and function g maps set B to set C then composition of f and g maps set A to set C . Mathematically, $f: A \rightarrow B \wedge g: B \rightarrow C \Rightarrow g \circ f: A \rightarrow C$

Properties

1. **Associativity:** Composition of functions is associative i.e. the order of grouping does not affect the composite function. Mathematically, $h \circ (g \circ f)(x) = (h \circ g) \circ f(x)$
2. **Commutativity:** Composition of functions is not commutative i.e. the order of function affects the composite function. Mathematically, $g \circ f(x) \neq f \circ g(x)$
3. **Injectivity:** If the functions f and g are injective functions then their composite function will also be an injective function.
4. **Surjectivity:** If the functions f and g are surjective functions then their composite function will also be a surjective function.

#Algebraic Function

An algebraic function is the function formed by a finite number of algebraic operations like addition, subtraction, multiplication, division and extraction of roots. E.g., $f(x) = 2x^2 - 5x$, $g(x) = \frac{x^2 - 1}{3x + 7}$, $h(x) = \sqrt{x + 3}$, etc.

Polynomial Function

A polynomial function is an algebraic function such that the power of the input variable is a whole number.

General form of polynomial function is given by

$$f(x) = \sum_{i=0}^n a_i x^{n-i} = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_{n-1} x^1 + a_n$$

Function	Description	General Form
Linear Function	Linear function is a polynomial function of 1st degree. Its graph is a straight line.	$f(x) = ax + b$ where a and b are constants.
	Identity Function: Identity function is the function which returns the same given input. Its graph is a straight line inclined at 45° from positive X-axis. It always passes through the origin.	$f(x) = x$
	Constant Function: Constant function is the function which always returns the same constant. Its graph is a straight line parallel to the X-axis.	$f(x) = k$ where k is any constant.
Quadratic Function	Quadratic function is a polynomial function of 2nd degree. Its graph is a parabola	$f(x) = ax^2 + bx + c, (a \neq 0)$ where a, b , and c are constants.
Cubic Function	Cubic function is a polynomial function of 3rd degree. Its graph is a curve symmetric about origin.	$f(x) = ax^3 + bx^2 + cx + d, (a \neq 0)$ where a, b, c , and d are constants.

#Transcendental Function

Transcendental means non-algebraic. Transcendental function is the function which cannot be expressed as the finite series of algebraic terms. This function can be expressed as the infinite series of algebraic terms.

Some of its types are mentioned below:

1. Trigonometric Function

Trigonometric function is the function of an angle defined as the ratio between the sides of a right angled triangle in a unit circle. This function is also called circular and cyclic function.

The fundamental trigonometric functions are:

Function	Definition	Domain	Range
Sine Function	The function of an angle defined as the ordinate of the point in a unit circle at the given angle is called sine function. Its general form is $f(\theta) = \sin \theta$.	$(-\infty, \infty)$	$[-1, 1]$
Cosine Function	The function of an angle defined as the abscissa of the point in a unit circle at the given angle is called cosine function. Its general form is $f(\theta) = \cos \theta$.		

2. Exponential Function

Exponential function is the function in which the input variable is the power of a constant.

Its general form is $f(x) = a^x$ where a is a positive real number.

Its graph is a growing curve.

3. Logarithmic Function

Logarithmic function is the inverse function of an exponential function.

Its general form is $f(x) = \log_a x$ where a is a positive real number.

Its graph is the reflection of its corresponding exponential function's graph about the line $y = x$.

Note:

- $\log_a b$ is read as 'Logarithm of b to the base a '
- Common Logarithm $\rightarrow$ Logarithm to the base 10
- Natural Logarithm $\rightarrow$ Logarithm to the base e
- We generally not write e in the base of logarithm.

$$\left. \begin{array}{l} \log x \\ \ln x \end{array} \right\} \log_e x$$

Properties of Logarithmic Function:

1. **Product Rule:** $\log_a(xy) = \log_a x + \log_a y$

2. **Quotient Rule:** $\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$

3. **Power Rule:** $\log_a(x^n) = n \log_a x$

4. **Change of Base Rule:** $\log_a b = \frac{\log_c b}{\log_c a}$

a. **Reciprocal Rule:** $\log_a b = \frac{1}{\log_b a}$ $[\because c \rightarrow b]$

b. **Chain Rule:** $\log_a b = \log_a c \cdot \log_c b$ $[\because \frac{1}{\log_c a} = \log_a c]$

4. Hyperbolic Function

Hyperbolic function is the function formed by the combination of exponential function e^x and e^{-x} . They are named hyperbolic functions because they are related to the conic section hyperbola.

Hyperbolic Function	Definition
Hyperbolic sine of x	$\sinh x = \frac{e^x - e^{-x}}{2}$
Hyperbolic cosine of x	$\cosh x = \frac{e^x + e^{-x}}{2}$
Hyperbolic tangent of x	$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$
Hyperbolic cotangent of x	$\coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}}$
Hyperbolic secant of x	$\operatorname{sech} x = \frac{2}{e^x + e^{-x}}$
Hyperbolic cosecant of x	$\operatorname{cosech} x = \frac{2}{e^x - e^{-x}}$

Basic Formulas of Hyperbolic Function

1. $\cosh^2 x - \sinh^2 x = 1$
2. $\tanh^2 x + \operatorname{sech}^2 x = 1$
3. $\coth^2 x - \operatorname{cosech}^2 x = 1$
4. $\sinh(-x) = -\sinh x$
5. $\cosh(-x) = \cosh x$
6. $\sinh 2x = 2 \sinh x \cosh x$
7. $\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$
8. $\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$
9. $\cosh(x - y) = \cosh x \cosh y - \sinh x \sinh y$